Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

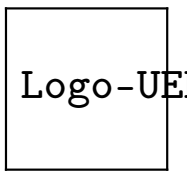
Ringkasan

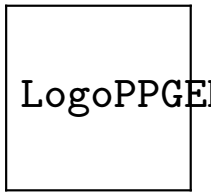
PENDAHULUAN
TINJAUAN PUSTAKA
ragged2e caption graphicx subfig tikz pgfplots spy,calc hyperref listings tikzpagenodes UERJpresentation

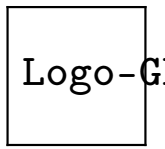
Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

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19 Pebruari 2025

Introduction

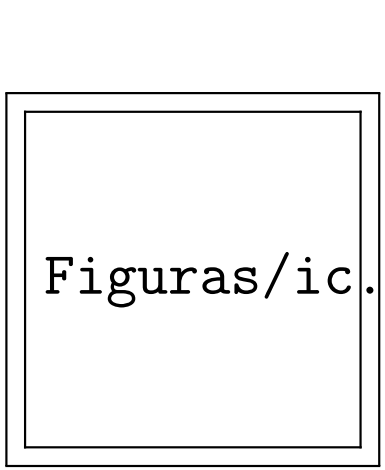
One of the biggest challenges on the design of modern electronic devices is the thermal control.

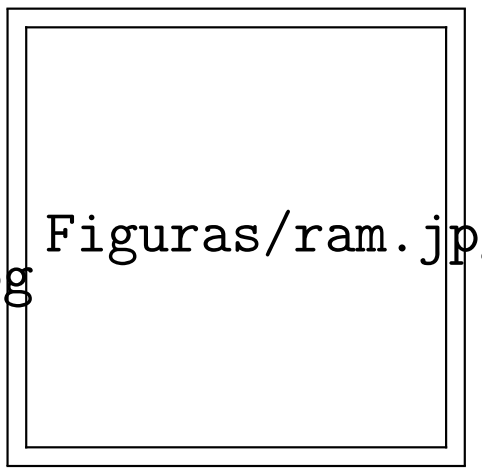
In this work, it is proposed a solution by Classical Integral Transform Technique(CITT) to solve a heat conduction problem on Solid-State Electronics (SSE) considering heat generation and external convection.

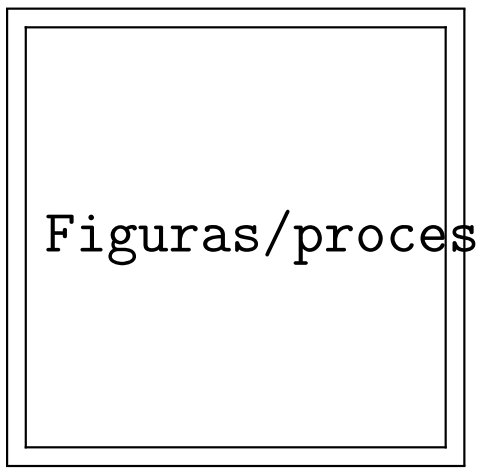
Solid State Electronics (SSE)

Solid-state electronics (SSE) are those circuits or devices built entirely from solid materials and in which the electrons, or other charge carriers, are encapsulated within the solid material.

Exemples: Transistors, Microprocessors, Random Access Memory (RAM), Itegrated Circuits (IC).







Substrate

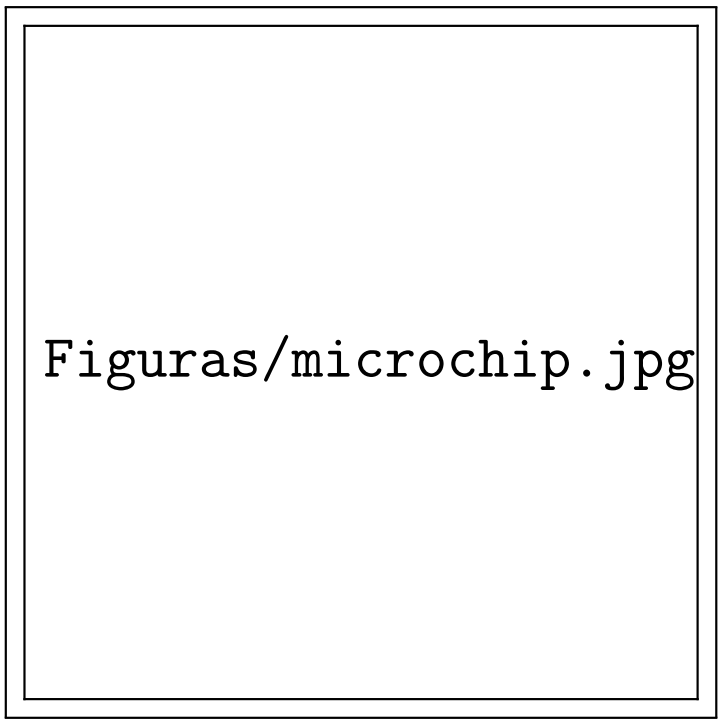
Substrate is a solid substance and serve as the foundation upon which electronic devices such as transistors, diodes, and especially Integrated Circuits (ICs) are encapsulated.

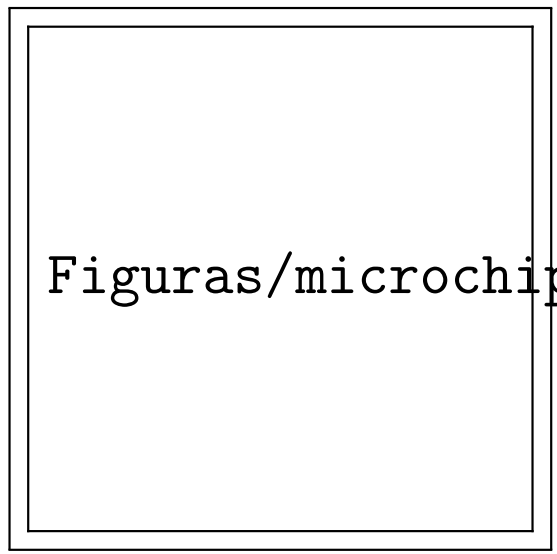
Example of substrate's materials:

- silicon
- silicon dioxide
- aluminum oxide
- sapphire

The advantage of this is the superior insulation between adjacent transistors.

Exemple of SSE: Integrated Circuit





Problem Description

Problem Formulation

The nondimensionalization of the problem leads to the following mathematical formulation:

$$\frac{\partial^2\Theta}{\partial\xi^2}+\beta^2\frac{\partial^2\Theta}{\partial\eta^2}-(\mathrm{Bi}\gamma)\Theta=-G(\xi,\eta)\quad\text{for}\quad 0\le\xi\le 1\quad\text{and}\quad 0\le\eta\le 1$$

$$\left.\frac{\partial\Theta}{\partial\xi}\right|_{\xi=0}=0;\qquad\left.\frac{\partial\Theta}{\partial\eta}\right|_{\eta=0}=0;$$

$$\left.\frac{\partial\Theta}{\partial\xi}\right|_{\xi=1}=0;\qquad\left.\frac{\partial\Theta}{\partial\eta}\right|_{\eta=1}=0;$$

Classical Integral Transform Technique

In order to solve the proposed problem, the Classical Integral Transform Technique (CITT) is applied, considering the appropriate Helmholtz Eigenvalue Problem in cartesian coordinates.

$$\Psi_n''(\eta) + \lambda_n^2 \Psi_n(\eta) = 0$$
$$\Psi_n'(0) = 0; \quad \Psi_n'(1) = 0.$$

where $\Psi(\eta)$ are the eigenfunctions and λ_n are the eigenvalues. For this particular problem, the case where $\lambda = 0$ also exists.
For $\lambda = 0$, the solution of the eigenvalue problem is given by:

$$\Psi_0(\eta) = 1; \qquad \lambda_0 = 0;$$

And for $\lambda > 0$:

$$\Psi_n(\eta) = \cos(\lambda_n \eta); \qquad \lambda_n = n\pi, \quad \text{for} \quad n = 1, 2, 3, \ldots$$

Classical Integral Transform Technique

To apply the CITT, the transformation pair is defined.
[10cm]Transformation Pair

$$\text{Transformation} \quad \Rightarrow \quad \bar{\Theta}_n(\xi) = \int_0^1 \Theta \Psi_n(\eta) \mathrm{d}\eta$$

$$\text{Inversion} \quad \Rightarrow \quad \Theta = \sum_{n=0}^{\infty} \frac{\bar{\Theta}_n(\xi) \Psi_n(\eta)}{N_n}$$

N_n is the norm and is defined as:

$$N_n = \int_0^1 \Psi_n^2 \mathrm{d}\eta$$

Classical Integral Transform Technique

The dimensionless diferential equation is written again, multiplied by Ψ_n and integrated in the domain for η.

$$\int_0^1 \frac{\partial^2 \Theta}{\partial \xi^2}\Psi_n d\eta + \beta^2 \int_0^1 \frac{\partial^2 \Theta}{\partial \eta^2}\Psi_n d\eta - \mathrm{Bi}\gamma \int_0^1 \Theta\Psi_n d\eta = -\int_0^1 G\Psi_n d\eta$$

The transformed equation is obtained and admits an analytical solution.

For $\lambda > 0$:

$$\bar{\Theta}_n'' + (-\beta^2\lambda_n^2 - \mathrm{Bi}\gamma)\bar{\Theta}_n = -\bar{G}_n(\xi)$$

$$\bar{\Theta}_n'(0) = 0; \quad \bar{\Theta}_n'(1) = 0$$

where

$$\bar{G}_n(\xi) = \int_0^1 G(\xi,\eta)\Psi_n(\eta)\mathrm{d}\eta$$

For $\lambda = 0$:

$$\bar{\Theta}_0'' - (\mathrm{Bi}\gamma)\bar{\Theta}_0 = -\bar{G}_0(\xi)$$

$$\bar{\Theta}_0'(0) = 0; \quad \bar{\Theta}_0'(1) = 0$$

where

$$\bar{G}_0(\xi) = \int_0^1 G(\xi,\eta)\mathrm{d}\eta$$

Classical Integral Transform Technique

The transformed equations admit a analytical solutions.

In order to obtain the final temperature field, the inversion formula must be utilized and the summation must be truncated to a finite value ($n_{\max}$).

$$\text{Inversion} \quad \Rightarrow \quad \Theta = \sum_{n=0}^{n_{\max}} \frac{\bar{\Theta}_n(\xi)\Psi_n(\eta)}{N_n}$$

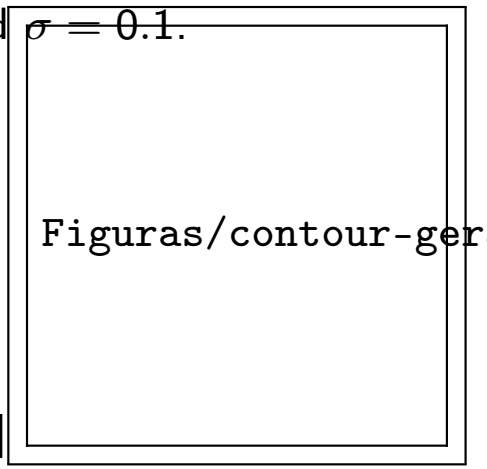
Heat Generation

$$G(\xi,\eta) = \frac{G_e}{\sigma^2\pi}\exp\left[-\frac{(\xi-\xi_0)^2+(\eta-\eta_0)^2}{\sigma^2}\right]$$

For this current work, two cases are considered:

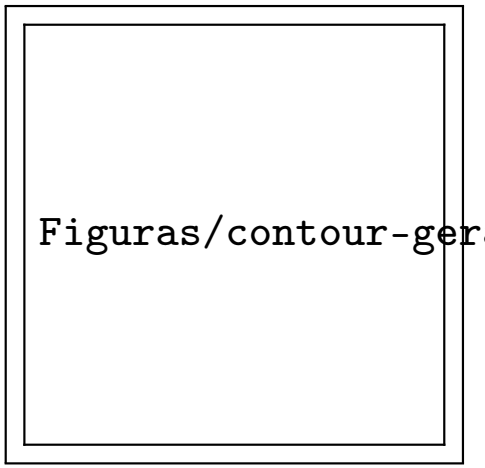
- Centered HG where $\xi_0 = 1/2$ and $\eta_0 = 1/2$
- Off-Centered HG where $\xi_0 = 1/4$ and $\eta_0 = 2/5$

For both cases $G_e = 1$ and $\sigma = 0.1$.



[][Centered HG]

[][Off-Centered



HG]

Results

Tabel: $\Theta(\xi, \eta)$ convergence solved for Centered HG by CITT

	Biγ=1 & β=1				Biγ=0.1 & β=1			
n_{max}	Θ(0.2,0.4)	Θ(0.5,0.5)	Θ(0.6, 0.8)	Θ(0.9,0.9)	Θ(0.2, 0.4)	Θ(0.5, 0.5)	Θ(0.6,0.8)	Θ(0.9,0.9)
1	0.981222	1.05643	1.03739	0.966733	9.98087	10.0575	10.0382	9.96593
5	1.00221	1.18948	0.997708	0.954075	**10.0025**	10.1922	9.99803	**9.95274**
10	**1.00213**	1.20298	1.00209	**0.954083**	10.0025	10.2058	10.0024	9.95274
20	1.00213	**1.20323**	**1.00213**	0.954083	10.0025	**10.2060**	**10.0025**	9.95274
30	1.00213	1.20323	1.00213	0.954083	10.0026	10.2060	10.0025	9.95274
	Biγ=1 & β=0.8				Biγ=0.1 & β=0.8			
n_{max}	Θ(0.2,0.4)	Θ(0.5,0.5)	Θ(0.6,0.8)	Θ(0.9,0.9)	Θ(0.2,0.4)	Θ(0.5,0.5)	Θ(0.6,0.8)	Θ(0.9,0.9)
1	0.981222	1.05643	1.03739	0.966733	9.98087	10.0575	10.0382	9.96593
5	1.01980	1.23495	0.980675	0.940388	10.0212	10.2395	9.98041	9.93805
10	**1.01957**	1.25429	0.987398	**0.940418**	**10.0210**	10.2589	9.98715	**9.93808**
20	1.01957	**1.25467**	**0.987461**	0.940418	10.0210	**10.2593**	**9.98722**	9.93808
30	1.01957	1.25467	0.987461	0.940418	10.0210	10.2593	9.98722	9.93808
	Biγ=1 & β=0.5				Biγ=0.1 & β=0.5			
n_{max}	Θ(0.2,0.4)	Θ(0.5,0.5)	Θ(0.6,0.8)	Θ(0.9,0.9)	Θ(0.2,0.4)	Θ(0.5,0.5)	Θ(0.6,0.8)	Θ(0.9,0.9)
1	0.981222	1.05643	1.03739	0.966733	9.98087	10.0575	10.0382	9.96593
5	1.09885	1.38536	0.921228	0.874113	10.1105	10.4039	9.91626	9.86204
10	1.09731	1.42441	0.936699	0.874431	**10.1090**	10.4432	9.93187	**9.86236**
20	**1.09730**	**1.42526**	**0.936864**	**0.874429**	10.1090	**10.4441**	**9.93203**	9.86236
30	1.09730	1.42526	0.936864	0.874429	10.1090	10.4441	9.93203	9.86236

Results

Tabel: $\Theta(\xi, \eta)$ convergence solved for Off-Centered HG by CITT

	Biγ=1 & β=1				Biγ=0.1 & β=1			
n_{max}	Θ(0.2,0.4)	Θ(0.5,0.5)	Θ(0.6,0.8)	Θ(0.9,0.9)	Θ(0.2,0.4)	Θ(0.5,0.5)	Θ(0.6,0.8)	Θ(0.9,0.9)
1	1.14595	0.991937	0.920318	0.865087	10.1555	9.99006	9.91129	9.84928
5	1.26434	1.02009	0.914914	**0.863013**	10.2754	10.0190	9.90570	**9.84707**
10	1.27061	**1.01987**	**0.91489**	0.863013	10.2817	**10.0188**	**9.90568**	9.84707
20	**1.27083**	1.01987	0.91489	0.863013	**10.2819**	10.0188	9.90568	9.84707
30	1.27083	1.01987	0.91489	0.863013	10.2819	10.0188	9.90568	9.84707
	Biγ=1 & β=0.8				Biγ=0.1 & β=0.8			
n_{max}	Θ(0.2,0.4)	Θ(0.5,0.5)	Θ(0.6,0.8)	Θ(0.9,0.9)	Θ(0.2,0.4)	Θ(0.5,0.5)	Θ(0.6,0.8)	Θ(0.9,0.9)
1	1.1564	0.991937	0.900772	0.846859	10.1682	9.99006	9.88666	9.82559
5	1.32297	1.04002	0.890144	**0.840922**	10.3380	10.0401	9.87552	**9.81914**
10	1.33221	**1.03950**	**0.890058**	0.840922	10.3473	**10.0396**	**9.87544**	9.81914
20	**1.33253**	1.03950	0.890058	0.840922	**10.3476**	10.0396	9.87544	9.81914
30	1.33253	1.03950	0.890058	0.840922	10.3476	10.0396	9.87544	9.81914
	Biγ=1 & β=0.5				Biγ=0.1 & β=0.5			
n_{max}	Θ(0.2,0.4)	Θ(0.5,0.5)	Θ(0.6,0.8)	Θ(0.9,0.9)	Θ(0.2,0.4)	Θ(0.5,0.5)	Θ(0.6,0.8)	Θ(0.9, 0.9)
1	1.18823	0.991937	0.830465	0.772371	10.2157	9.99006	9.77686	9.70579
5	1.52152	1.12367	0.794684	0.738850	10.5654	10.1341	9.73726	9.66629
10	1.54154	1.12104	0.793933	**0.738828**	10.5855	10.1315	9.73650	**9.66627**
20	1.54228	**1.12102**	**0.793928**	0.738828	**10.5862**	**10.1314**	**9.73649**	9.66627
30	**1.54229**	1.12102	0.793928	0.738828	10.5862	10.1314	9.73649	9.66627

Results

Gambar: Illustration of the solution by CITT with the different heat generation positions with the parameters of $\mathrm{Bi}\gamma$=1 and β=1.

Figuras/contour-citt-assimetrico.pdf

[][Off-Centered HG case]

[][Centered

Figuras/contour-citt-central.pdf

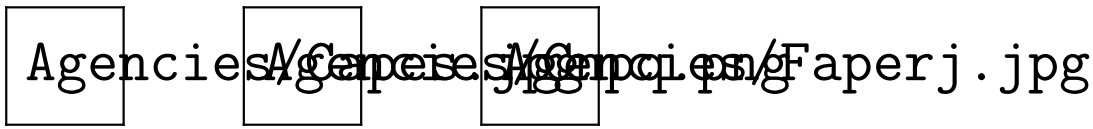
HG case]

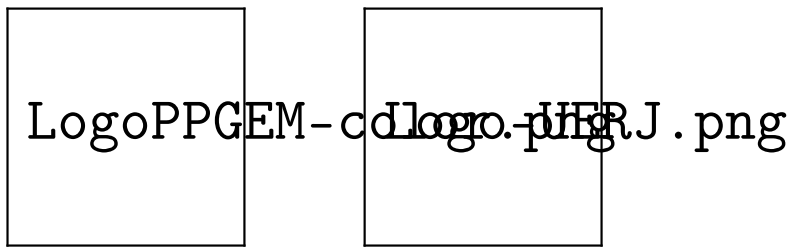
Conclusions

- The solution is fully analytical
- CITT had a great performance to obtain high accuracy with very few terms in the solution summation.
- This is a preliminary work and for near future works, one can propose: Use of the complete Heat Generation and variation of the heat conductivity.

Thank You

Acknowledgments:





Lists - Examples

- Point A
- Point B
  - part 1
  - part 2
- Point C

I Point A
II Point B
  i part 1
  ii part 2
III Point C

API Application Programming Interface
LAN Local Area Network
ASCII American Standard Code for Information Interchange

Block Texts - Examples

Block Title

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod

Block Title

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor

Definition

A prime number is a number that...

Example

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Including Code - Examples

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \tag{1}$$